

MedAxis—National Prescription Trust Infrastructure - Technical Documentation Template

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Link to Github repo with all docs and diagrams: [Github](#)

1. Executive Summary

In Morocco and across Africa, the prescription system is broken in ways that are invisible until something goes wrong. A doctor writes a prescription on paper. That paper can be forged, reused, or never filled at all. A pharmacist has no way to confirm the doctor is real, licensed, or even still practicing. A patient pays full price upfront and waits weeks for insurance to reimburse them, if it ever does. A regulator sees none of this in real time.

MedAxis is a national prescription trust infrastructure that fixes this by anchoring every prescription to a verified digital identity. When a doctor issues a prescription through MedAxis, their Ministry of Health license is cryptographically verified through the MOSIP eSignet and Inji Certify stack. The prescription is stored as a metadata record in a national ledger linked directly to the patient's national ID. No clinical notes, no biometrics, and no full medical records are ever stored centrally.

When the patient walks into a pharmacy, the pharmacist enters the patient's CNIE number. The system retrieves all active prescriptions linked to that identity, verifies the prescribing doctor's license in real time, confirms the patient through a simple OTP, and only then allows dispensing. The prescription is marked as dispensed and can never be reused.

Insurance claims are validated automatically against the ledger, eliminating paper submissions and reimbursement delays entirely.

The core value of MedAxis is not that it digitizes prescriptions. It is that it makes prescription fraud structurally impossible, protects patient privacy by design, and creates the first tamper-proof national audit trail for pharmaceutical dispensing in Morocco, with a clear path to scale across the African continent.

2. User Stories (For detailed acceptance criteria for each user story, refer to Appendix A)

ACTOR 1: DOCTOR

US-D01 As a licensed doctor, I want to authenticate into MedAxis using my national CNIE ID via eSignet so that my identity is cryptographically verified before I can issue any prescription.

US-D02 As a doctor, I want my Ministry of Health medical license to exist as a verifiable credential in my Inji Wallet so that pharmacies can verify my licensing status cryptographically without calling the Ministry.

US-D03 As a doctor, I want to create a digital prescription by filling out a form with the patient ID, drug code, dosage, frequency, and duration so that the prescription is automatically stored in the national ledger and linked to the patient's digital ID.

US-D04 As a doctor, I want to search for a patient by their national ID before issuing a prescription so that I confirm I am prescribing to the correct person.

US-D05 As a doctor, I want all prescriptions I issue to automatically expire after the duration I specify so that old or unused prescriptions cannot be misused after the treatment window.

US-D06 As a doctor, I want to view a history of all prescriptions I have issued, filtered by date, patient token, drug code, or status, so that I can track my prescribing activity and spot anomalies.

US-D07 As a doctor, I want to cancel a prescription I issued before it is dispensed so that I can correct mistakes without creating fraudulent records.

ACTOR 2: PHARMACIST

US-P01 As a pharmacist, I want to authenticate into MedAxis using my national CNIE ID via eSignet so that my identity is verified before I can dispense any medication.

US-P02 As a pharmacist, I want to enter a patient's national ID to retrieve all their active prescriptions so that I can identify the correct prescription to dispense without requiring a QR code or paper document.

US-P03 As a pharmacist, I want the system to automatically verify the prescribing doctor's license (verifiable credential) before I can dispense so that I never dispense on behalf of an unlicensed or suspended doctor.

US-P04 As a pharmacist, I want to verify the patient's identity via OTP before dispensing so that medication cannot be collected by an unauthorized person.

US-P05 As a pharmacist, I want to see a clear summary of all three verification checks before I confirm dispensing so that I have a single confirmation screen showing doctor licensed, prescription valid, and patient verified.

US-P06 As a pharmacist, I want to record a dispensing event by confirming the exact quantity dispensed so that partial dispensing is supported when full stock is unavailable.

US-P07 As a pharmacist, I want to view my pharmacy's dispensing history by date and drug code so that I can run stock reconciliation and audit my dispensing activity.

US-P08 As a pharmacist, I want to flag a suspicious prescription for regulator review without blocking dispensing so that I can report anomalies while still serving the patient.

ACTOR 3: PATIENT

US-PT01 As a patient, I want to view a read-only history of all prescriptions issued in my name so that I can detect any prescriptions I did not authorize.

US-PT02 As a patient without a smartphone, I want a pharmacist to verify my identity using an OTP sent to my basic feature phone so that I am not excluded from the system due to lack of technology.

US-PT03 As a patient, I want to receive an SMS notification when a prescription issued in my name is dispensed so that I am immediately alerted if an unauthorized dispensing occurs.

US-PT04 As a patient, I want to report an unauthorized prescription or dispensing event directly from my patient portal so that I can trigger an investigation without going to a physical office.

ACTOR 4: REGULATOR (Ministry of Health)

US-R01 As a Ministry of Health official, I want a real-time dashboard showing all prescription and dispensing activity nationwide so that I can monitor the health of the prescription system at a glance.

US-R02 As a regulator, I want to view a full audit trail for any specific prescription by RxID so that I can reconstruct exactly what happened from issuance to dispensing.

US-R03 As a regulator, I want to activate, suspend, or revoke a doctor's license in the system so that suspended doctors are immediately blocked from issuing valid prescriptions.

US-R04 As a regulator, I want to see a ranked list of doctors by prescription volume for controlled substances so that I can identify over-prescribing patterns early.

US-R05 As a regulator, I want to register new pharmacies and assign pharmacist accounts so that only licensed dispensing points can operate on the platform.

ACTOR 5: INSURER (e.g. Ma MGPAP, NHIF, NHIS)

US-I01 As an insurance provider, I want every reimbursement claim to be automatically matched to a verified, single-use prescription in the MedAxis ledger so that I can approve or reject claims programmatically without manual review.

US-I02 As an insurer, I want to query dispensing history for a specific patient token so that I can detect patients submitting duplicate claims across multiple insurers.

US-I03 As an insurer, I want a monthly report of all claims processed through MedAxis for my patient pool so that I can reconcile with my internal accounting system.

ACTOR 6: SYSTEM / ADMIN

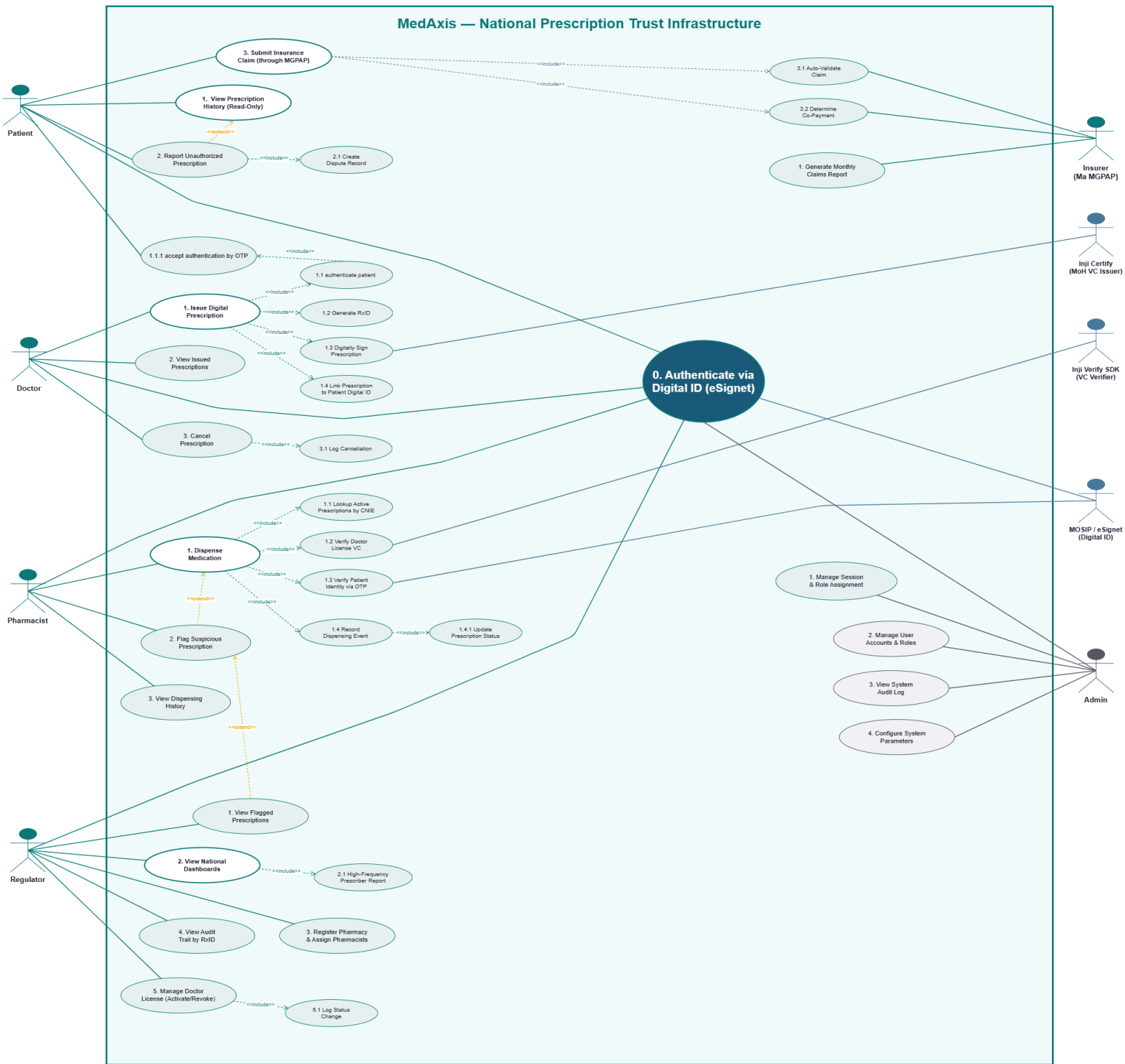
US-SYS01 As a system administrator, I want all API calls to be logged with actor ID, endpoint, timestamp, request hash, and response status so that there is a complete immutable audit trail for legal and compliance purposes.

US-SYS02 As a system administrator, I want role-based access control enforced at the API level so that no actor can access endpoints outside their permitted role.

US-SYS03 As a system administrator, I want the system to reject any prescription dispensing attempt if eSignet authentication has expired so that session hijacking cannot be used to fraudulently dispense medication.

3. System Use Cases

- Use Case Diagram:



Link for a clearer view of the use case diagram: [Use case diagram Link](#)

User Authentication: Every actor in MedAxis—doctor, pharmacist, patient, regulator, and admin—authenticates through MOSIP eSignet using their national CNIE ID before accessing any system function. The system receives a signed JWT token confirming verified identity and assigns the appropriate role-based session.

Prescription Issuance: An authenticated doctor searches for a patient by national ID, fills a prescription form with drug code, dosage, frequency, and duration, and submits it. The system generates a unique RxID, digitally signs the prescription using Inji Certify, links it to the patient's Digital ID in the national ledger, and confirms issuance to the doctor.

Prescription Retrieval and Dispensing: A pharmacist enters the patient's CNIE number to retrieve all active prescriptions linked to that identity. The system automatically verifies the prescribing doctor's license through the Inji Verify SDK, sends an OTP to the patient's phone for identity confirmation, and presents a three-check verification screen. Only when all three checks pass does the system allow the pharmacist to confirm dispensing and update the prescription status in the ledger.

Patient Transparency: An authenticated patient accesses a read-only portal showing all prescriptions ever issued in their name. They can view prescription status and report any unauthorized prescription or dispensing event, which creates a disputed record visible to regulators.

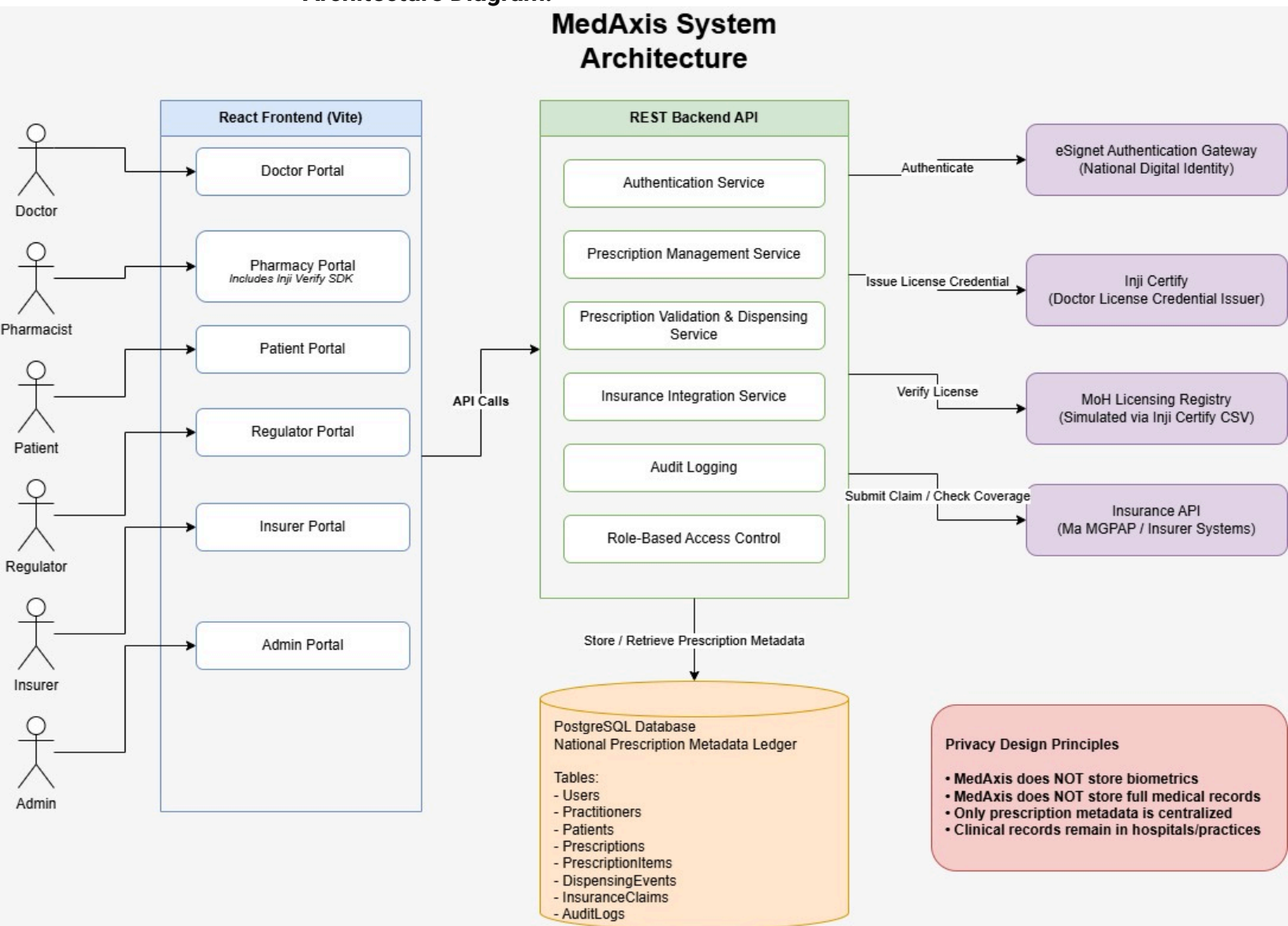
Regulatory Oversight: An authenticated regulator accesses a real-time national dashboard showing prescription and dispensing activity across the country. They can view full audit trails by RxID, manage doctor license statuses, register pharmacies, and monitor high-frequency prescribing patterns for controlled substances.

Insurance Claim Processing: An insurer submits a reimbursement claim referencing a specific RxID. The system automatically checks that the prescription was dispensed, that the doctor license was verified at dispensing time, and that no duplicate claim exists. Claims are approved or rejected programmatically within two seconds.

System Administration: An admin manages user accounts, assigns roles, configures system parameters, and accesses the full append-only audit log of every API call made across the platform.

4. System Architecture

- Architecture Diagram:

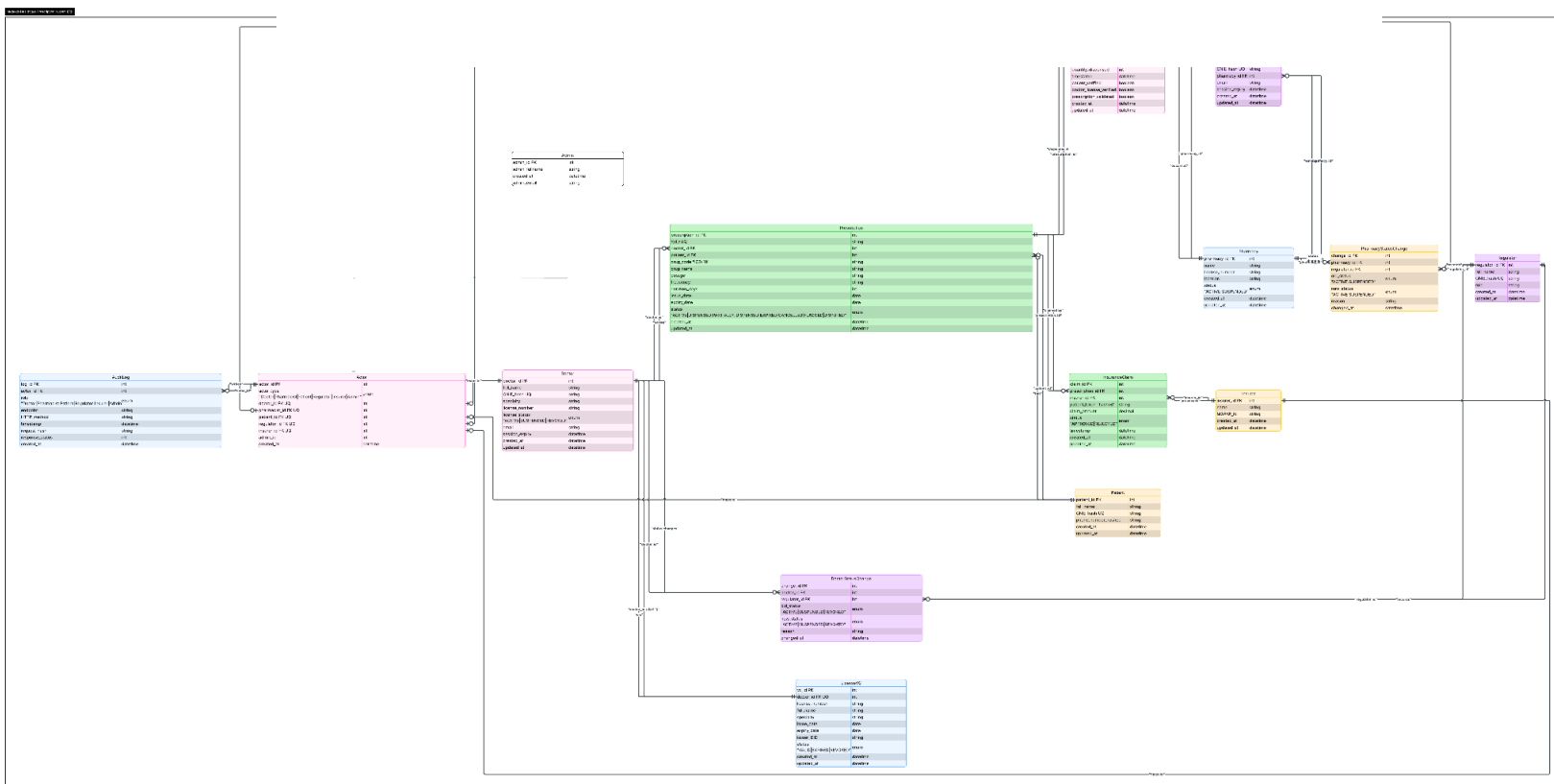


Link for a clearer view of the diagram: [System architecture diagram](#)

- Tech Stack:

- **Frontend:** React (Vite)
- **Backend:** Node.js, Express, Prisma ORM,
- **Database:** PostgreSQL
- **Infrastructure:** Docker (eSignet + Inji Certify local sandbox), Railway (backend hosting)

5. Data Model & Design



Entity Relationship Diagram (ERD): [download the PDF file for better view & zoom:](#)

[Link 1 Github](#)

[Link 2 Lucid chart](#) (in case the 1st link doesn't work)

- Key Tooling:

Prisma ORM: defines and manages all database models and generates the SQLite schema automatically from code.

SQLite: a lightweight file-based relational database storing all prescription metadata, users, dispensing events, audit logs, and insurance claims.

MOSIP eSignet: verified identity tokens from eSignet populate actor identities across all database tables.

Inji Certify: issues doctor licenses with verifiable credentials that populate the LicenseVC table using a Ministry of Health CSV data source.

Inji Verify SDK: cryptographically verifies doctor license VCs at dispensing time; the result is stored in the DispensingEvent table as the doctor_license_verified boolean.

APPENDIX A—Acceptance Criteria

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US-D01

- Doctor cannot access prescription issuance without completing eSignet OIDC authentication
- The system receives and validates signed JWT from eSignet
- Failed authentication returns clear error message
- Session expires after 8 hours of inactivity

US-D02

- License VC is issued by Ministry of Health via Inji Certify
- VC contains license number, full name, specialty, issue date, expiry date, issuer DID
- Doctor can view their VC in Inji Wallet app
- Expired or revoked VCs are rejected at verification time

US-D03

- The form requires a patient national ID token, ICD-10 drug code, dosage, frequency, duration in days
- The system generates a unique RxID (UUID)
- The prescription is automatically linked to the patient's national ID token in the ledger
- The prescription is stored with status Dispensed=FALSE and calculated expiry date
- The doctor receives confirmation with RxID and prescription summary
- No QR code is generated—the patient presents their CNIE at the pharmacy

US-D04

- Doctor enters patient national ID; system returns anonymized patient token and first name only
- No full patient record is returned—minimal data only
- If patient does not exist in system, doctor can initiate patient registration

US-D05

- Expiry date = issue date + duration specified by doctor
- The system enforces a maximum expiry window of 90 days
- Expired prescriptions return status EXPIRED and cannot be dispensed
- Expiry is checked at the moment of dispensing, not at issuance

US-D06

- The doctor sees list of own prescriptions only—they cannot view other doctors' prescriptions
- Filterable by date range, drug code, status (active / dispensed / expired)
- Each row shows RxID, patient token (masked), drug, issued date, status
- No clinical notes or full patient PII visible

US-D07

- Cancellation only allowed if Dispensed=FALSE and the prescription is not expired
- Cancelled prescription status changes to CANCELLED—not deleted
- Cancellation is logged with timestamp and doctor ID
- Cancelled prescriptions cannot be reinstated

US-P01

- Same eSignet OIDC flow as doctors
- The pharmacist role is assigned separately from doctor role at registration
- Failed auth blocks all dispensing actions

US-P02

- The pharmacist enters the patient's CNIE number into pharmacy portal
- The system returns all active, non-expired, non-dispensed prescriptions linked to that patient token
- Each prescription shows the drug name, drug code, dosage, frequency, prescribing doctor's name, issue date, expiry date
- The system does NOT return full patient PII—it returns patient token and first name only
- If no active prescriptions found, system returns clear NO ACTIVE PRESCRIPTIONS message
- The patient does not need a smartphone or printed document—CNIE is sufficient

US-P03

- Inji Verify SDK checks doctor's VC cryptographically against Ministry of Health public key
- Check happens automatically on prescription retrieval—the pharmacist does not trigger it manually
- If the VC is invalid, expired, or revoked: dispensing is blocked with clear reason shown
- The verification result is logged with a timestamp.

US-P04

- The pharmacist enters patient's phone number or national ID; system sends OTP to registered phone
- OTP is valid for 5 minutes only
- The patient confirms OTP verbally or by showing the screen to the pharmacist.
- If OTP fails 3 times, dispensing is blocked and event is logged
- Works on feature phones (SMS OTP)—does not require smartphone

US-P05

- The confirmation screen shows three checkmarks: Doctor VC Valid / Prescription Unused & Not Expired / Patient Identity Verified
- All three must be green before Confirm Dispense button is active
- Pharmacist cannot bypass any check

US-P06

- The pharmacist inputs quantity dispensed (up to prescribed quantity)
- If partial: prescription status changes to PARTIALLY_DISPENSED; remaining quantity tracked
- If full: prescription status changes to DISPENSED
- A second pharmacist at a different pharmacy can dispense the remainder on the same RxID

US-P07

- Filtered by date range, drug code, doctor ID
- Shows: RxID, drug, quantity, dispensing timestamp, patient token (masked)
- Cannot see other pharmacies' dispensing events
- Exportable as CSV for stock management

US-P08

- Flag button available on confirmation screen
 - Flagging adds a FLAGGED tag to the prescription record
 - The regulator is notified automatically
 - Dispensing can still proceed after flagging if all checks pass
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US-PT01

- The patient authenticates via eSignet (own CNIE)
- Sees: RxID (masked), drug name, issuing doctor name, issue date, status
- Cannot see dosage notes or clinical context—drug name only
- Records are read-only—no editing or deletion

US-PT02

- Assisted verification: pharmacist enters patient's phone number
- The system sends SMS OTP to feature phone (no app required)
- The patient reads OTP aloud to the pharmacist, who enters it or shows him phone screen
- This path is functionally equivalent to self-verification—same security guarantees

US-PT03

- SMS sent to patient's registered phone number on every dispensing event
- The message includes the drug name, pharmacy name, timestamp
- The patient cannot opt out of security notifications (only marketing opt-out allowed)
- If phone number not registered, notification is skipped silently and logged

US-PT04

- Report button available on each prescription record
 - Patient selects reason: Did not visit this doctor / Did not collect this medication / Other
 - The report creates a DISPUTED record visible to regulators
 - The patient receives confirmation reference number
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US-R01

- The dashboard shows total prescriptions today/this week/this month, dispensing rate, expired unused rate, flagged prescriptions count
- Filterable by: region, drug code, doctor specialty, date range
- Data updates in real time (or near real time with max 5-minute delay)
- Accessible via admin panel—no separate build required

US-R02

- The audit trail shows the issuance event, all verification checks, dispensing event, any flags or disputes
- Each event has an actor ID, timestamp, action, result
- Cannot be edited or deleted—append-only log

US-R03

- The regulator updates the doctor's status to ACTIVE / SUSPENDED / REVOKED
- Change takes effect immediately—no grace period for REVOKED
- All in-flight prescriptions from a REVOKED doctor are flagged automatically
- Action is logged with regulator ID and reason

US-R04

- List filterable by drug class (controlled / antibiotic / standard)
- Sortable by: total prescriptions, prescriptions per unique patient, prescriptions per week
- Threshold alerts: auto-flag doctors exceeding configurable threshold
- Data is anonymized for patients—doctor data is not anonymized

US-R05

- The regulator creates pharmacy record with the name, license number, location, assigned pharmacists
- Pharmacist accounts linked to a pharmacy before they can dispense
- Pharmacy can be suspended (all dispensing blocked) with one action

US-I01

- The insurer calls POST /api/insurance/claims/ with RxID and claim amount

- System checks: Dispensed=TRUE, VC verified at time of dispensing, claim not already submitted
- Auto-approved if all checks pass; auto-rejected with reason if any check fails
- Response time under 2 seconds

US-I02

- Insurer queries by patient token (hashed—no PII)
- Returns: list of dispensing events with RxID, drug code, pharmacy ID, timestamp
- Does not return drug names or clinical context—metadata only
- Rate-limited to prevent bulk harvesting

US-I03

- Report available as API endpoint and downloadable CSV
 - Contains: RxID, claim amount, approval status, timestamp
 - Patient token included but no PII
 - The report covers exactly one calendar month per request
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US-SYS01

- Every API call logged to append-only audit log table
- Logs include actor_id, role, endpoint, HTTP method, timestamp, response_status
- Logs cannot be edited or deleted via any API endpoint
- Log retention minimum 7 years (healthcare compliance)

US-SYS02

- Roles: DOCTOR, PHARMACIST, PATIENT, REGULATOR, INSURER, ADMIN
- Each endpoint has an explicit allowed roles list
- Unauthorized access returns HTTP 403 with no data leak
- Role assignment happens at account creation and can only be changed by ADMIN

US-SYS03

- eSignet token expiry checked on every dispensing action—not just on login
- An expired token forces re-authentication before proceeding
- Attempted dispensing with expired token is logged as a security event